



High Amplitude Lift Tracking Using Closed-Loop Feedback and Control

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Abstract

Two closed-loop feedback controllers based on PID and on full state-feedback were developed for high amplitude coefficient of lift tracking to expand the control capability of a pitching flat plate beyond the point of attached flow. All experiments were conducted at the University of Dayton Water Tunnel (UDWT) at a Reynolds number of 54,000. The two control architectures provided the capability to achieve C_L tracking at various C_L amplitudes well beyond stall and at reduced frequencies of up to 0.3. The angle of attack variation commanded by the controllers to track C_L at various amplitudes and frequencies showed different trends between the PID controller and the full-state feedback controller.

Nomenclature

α	= geometric angle of attack (radians)	f	= frequency (Hz)
α_d	= desired geometric angle of attack (rad)	t	= time (s)
c	= chord	τ	= convective time
C_L	= lift force coefficient		
s	= wing positing with respect to gust		
U_∞	= freestream velocity (m/s)		
$\dot{\alpha}$	= pitch rate (rad/s)		
$\ddot{\alpha}$	= pitch acceleration (rad/s ²)		
b	= semi-chord (m)		
a	= pitch axis location in semi-chord (m)		
$C(k)$	= Theodorsen's function		
$\dot{h}$	= plunge rate (m/s)		
$\ddot{h}$	= plunge acceleration (m/s ²)		
k	= reduced frequency		
ω	= angular frequency (rad/s)		

Abbreviations

GR	= Gust Ratio
PID	= Proportional Integral Derivative
LPF	= Low Pass Filter
SF	= State Feedback
FSF	= Full State Feedback

I. Introduction

Many studies in the fluid dynamics community have focused on the response of a pitching wing or airfoil in an open loop maneuver or during a gust encounter. These studies have unveiled the underlying physical complexities of operating fixed wing vehicles in unsteady environments. As there is a rise in the use of particularly small, fixed wing vehicles such as small unpiloted aerial vehicles (UAVs) which are susceptible to the effects of operating in

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unsteady environments, knowledge of how to control an aircraft through such encounters is of great interest. Another application of this research can be used in the controls for a fixed wing vertical take-off and landing vehicle where transition between vertical and forward flight presents a scenario where a controller is needed to maintain stability of the vehicle [1].

The work of Theodorsen [2] in understanding the lift response of an airfoil undergoing periodic pitching and plunging motion in an inviscid, incompressible flow, has been a fundamental model for understanding a wing's response to pitching maneuvers. It is expressed for the coefficient of lift as Eq. (1)

$$C_l = \pi b \left(\frac{\dot{\alpha}}{U_\infty} + \frac{\ddot{h}}{U_\infty^2} - \frac{ba\ddot{\alpha}}{U_\infty^2} \right) + 2\pi C(k) \left[\frac{\dot{h}}{U_\infty} + \alpha + b \left(\frac{1}{2} - a \right) \frac{\dot{\alpha}}{U_\infty} \right]. \quad (1)$$

This model has been used for comparisons to most control methods aimed at cases where the reduced frequency ($k = \frac{\omega c}{2U_\infty}$) and the amplitude of the motion where the flow is relatively attached and doesn't experience massive flow separation. Thakor [3] et al. carried out a computational study of a pitching SD7003 airfoil at low Reynolds number with comparisons to the Theodorsen model to show the influences of circulatory and non-circulatory lift components. This work concluded that the model compared well to the computational results for lower amplitude ($\alpha=2$ degrees), high reduced frequency cases. However, for higher amplitude ($\alpha = 6$ degrees) cases, experimental results were required to accurately calculate the coefficient of lift. For higher amplitude and frequency pitching however, a more accurate model is needed.

Brunton et al [4] present a modified version of the G-K model [5] in their work that is able to achieve better alignment with experimental data of a pitch oscillating airfoil but still requires large modification of the time constants. They were also able to show through a direct numerical simulation (DNS) study that the forces experienced by a low aspect ratio flat plate in a pitch up maneuver closely matched the forces on an impulsively started 2D plate, which suggests their proposed low order model impulsively started flat plates could be extended to these 3D cases. The results of one of their comparisons is shown in Fig. 1, where three static lift curve runs are compared to an oscillating pitch profile centered at 19 degrees angle of attack with a ± 4 degree amplitude at 0.5 Hz. When modifying the G-K model by a factor of 4, the resulting 2 Hz profile matches the measured lift response.

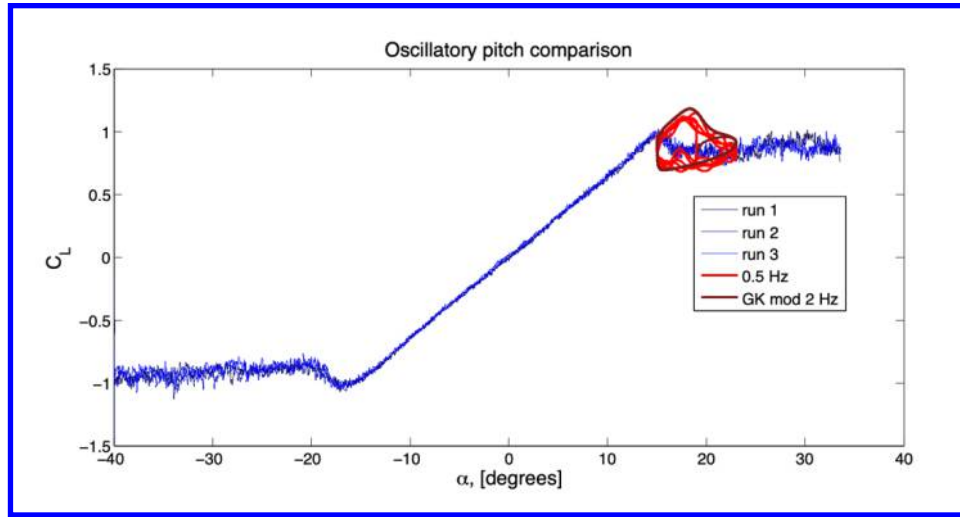


Fig. 1 Brunton [4] Static runs (1-3) compared to sinusoidal oscillations and G-K predicted lift.

Another approach carried out by An et al. [6] provides a systematic way for the G-K model to be used with a controller utilizing surface pressure readings. These readings are fed into a Kalman filter to give an estimate of the lift on a pitching wing and used as a state in a modified G-K model. The results of their model to data comparison are shown in Fig. 2. This provides great tracking to a random pitching movement at high angles of attack as well as a methodology for deploying this type of controller on a wing with surface pressure readings rather than force balance sensors.

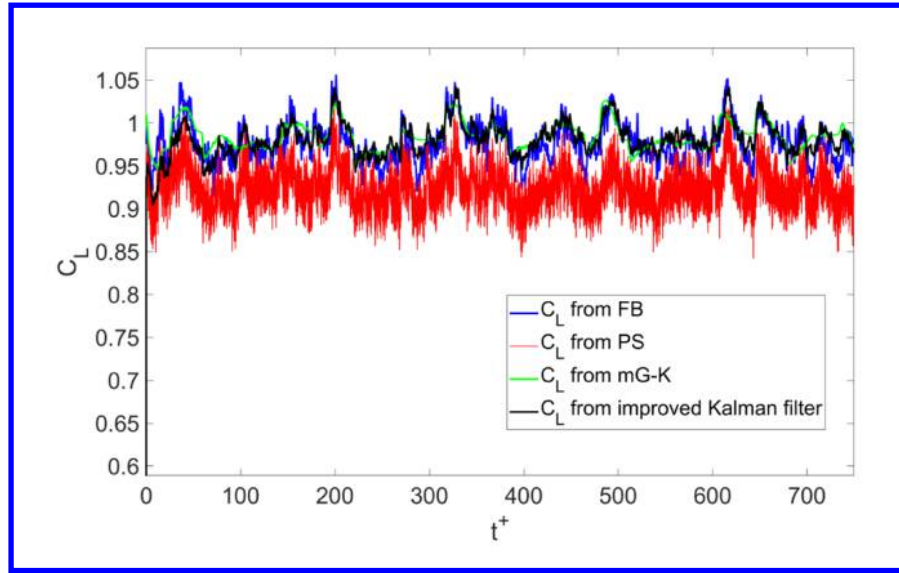


Fig. 2 An 2020 [6] Force Balance, Pressure Sensor, G-K model, and Improved G-K model comparison.

A recent study by Mongin and Gunasekaran [7] showed that an experimental closed loop system implementing a proportional, integral, derivative (PID) based control for coefficient of lift tracking proved accurate and stable at low reduced frequencies and low coefficient of lift amplitudes. However, this controller became unstable when higher amplitude desired coefficients of lift were commanded due to the non-linear behavior of the system beyond the static stall angle for the wing that was tested. The study also showed that as the reduced frequency of the desired tracking coefficient of lift was increased, the system's phase lag increased significantly. Each of these findings led to the need for a more robust controller that could take into account the physics of the wing in unsteady and highly unsteady reduced frequency regimes as well as at higher angles of attack wherein higher coefficients of lift are possible. The analysis done in this study looked at the simplified Theodorsen model [8] for the angle of attack history that was produced by the controller when tracking coefficient of lift. This model is shown in Equation 2.

$$C_l = 2\pi (F[1 + ik] + G[i - k])\alpha e^{i\omega t} + \pi k \left(i - \frac{k}{2}\right) \alpha e^{i\omega t} \quad (2)$$

The objective behind the study done by Mongin and Gunasekaran [7] was to find the point at which the Theodorsen model deviated greatly from the experimental results and to inform a future study of a feasible range to use the model. Figure 3 shows the plots of two reduced frequencies used for the study where the experimental closed loop control angle of attack history is plotted alongside the Theodorsen predicted angle of attack as found from the desired coefficient of lift profile of the system. The coefficient of lift history is also plotted, comparing the realized coefficient of lift of the closed loop system to that of the desired coefficient of lift profile. It can be clearly seen that at low reduced frequency, the closed loop control system was effectively able to track the desired coefficient of lift input, however, at higher reduced frequencies, significant differences in amplitude and phase can be seen. The current study is aimed at addressing this issue with a more robust PID and full-state feedback controllers.

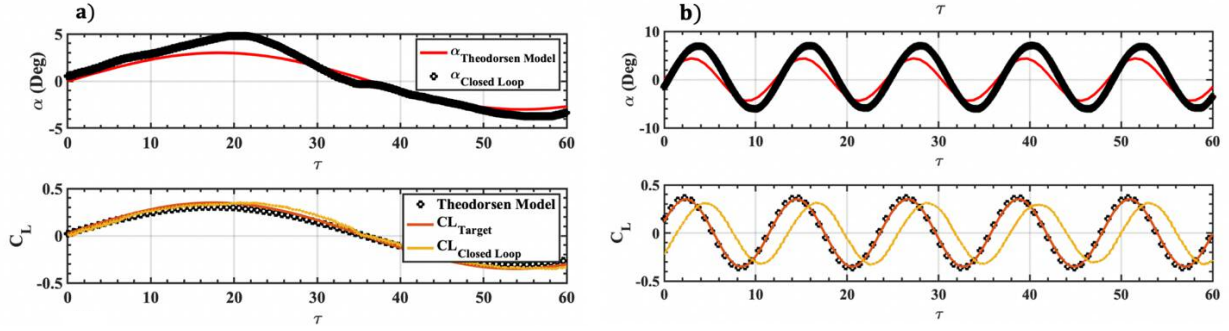


Fig. 3 Mongin and Gunasekaran [7] Comparison of angle of attack variation between the closed loop control and the Theodorsen model at a) reduced frequency of 0.04 and b) reduced frequency of 0.3.

In the current study, the pitch axis of a flat plate is commanded with a closed loop controller that tracks a desired coefficient of lift profile. The goal is to employ a control methodology that can accurately maintain tracking to a coefficient of lift profile that extends the plate well outside of the static stall lift response, allowing for a larger range of instantaneous lift capability. Use of PID and full state feedback controllers were implemented for this study to assess whether or not they are suitable for the application of tracking lift at higher amplitudes.

In full state feedback control, a set of gains is calculated using pole placement to achieve the desired system performance. This is done by first finding a model of the open loop plant which can be done via different means but in the present study, it was done with system identification wherein a step function was applied to the open loop system and the response was measured. The system is identified from this input-output relationship using MATLAB's state space estimation function. The results of this are given in state space form as matrices A, B, C, and D as shown in Fig. 4 and Equation 3. The full state feedback system incorporates a state vector gain K of which the product is subtracted from the reference signal r , yielding the controller input u . The final result is represented by the output y .

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}\quad (3)$$

Once an open loop model of the system is obtained and compared to the original data, the system can be trusted to represent the system and full state feedback control can be performed. The poles of the open loop system are found from the roots of the characteristic equation shown in Equation 4 in the Laplace domain.

$$|sI - A| = 0 \quad (4)$$

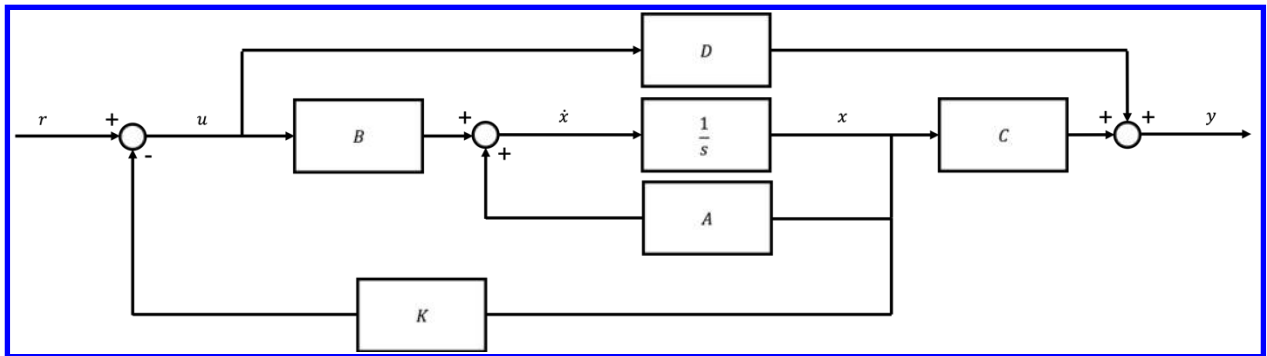


Fig. 4 Full State Feedback System

Where s is the Laplace variable, and I is the identity matrix. These can be found with the MATLAB 'roots' function as well. The desired poles of the system can be calculated from the roots of a second order system transfer function in terms of natural frequency and damping as shown in Equation 5 where ω_0 is the natural frequency and ζ is the damping ratio. This allows for modification of the response of the system in terms of useful parameters. Using the input u as the product of the state vector and the feedback gain K Equation 6 we obtain the closed loop system Equation 7.

$$TF = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2} \quad (5)$$

$$u = -Kx \quad (6)$$

$$\begin{aligned} \dot{x} &= (A - BK)x \\ y &= (C - DK)x \end{aligned} \quad (7)$$

The gains of the closed loop system can then be calculated by finding the roots of the determinate of the characteristic equation with the new closed loop relationship as shown in Equation 8 and comparing to the roots of the desired poles. This can also be achieved with the place function in MATLAB using the open loop A and B matrices and the desired poles.

$$\det[sI - (A - BK)] = 0 \quad (8)$$

These gains can then be implemented into the full state feedback controller and the system can be operated accordingly.

II. Experimental Setup

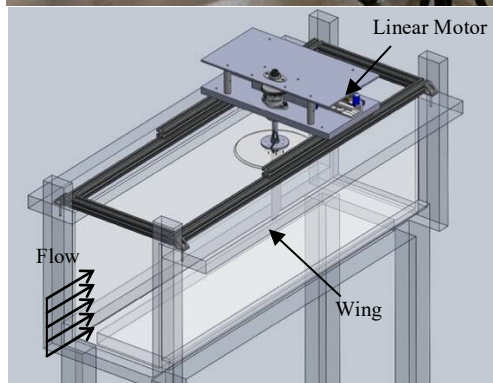
All experiments were performed at the University of Dayton Water Tunnel, shown in Fig. 5. For a detailed explanation of the experimental setup refer to Mongin and Gunasekaran [7]. The facility is a horizontal free-surface water tunnel with a 6:1 contraction and a 38 cm wide and 46 cm high test section of 152 cm length. The free-surface speed range is 7.5 cm/s to 38 cm/s. For this study, the freestream velocity of the tunnel was set to 0.371 m/s and a flat plate with a 5-inch (0.127m) chord and 10-inch (0.254m) span was used. The tunnel is equipped with a one degree of freedom motion stage consisting of a H2W Technologies linear motor driven by a Xenus XTL-230-18 motor driver from Copley Controls. This motor is outfitted with a rack and pinion gear that converts its linear motion to a rotational motion in order to pitch a test article in the tunnel test section. The closed-loop feedback and control for this system is performed with a dSPACE MicroLabBox with BNC type terminations using the MATLAB LabVIEW Real Time Interface (RTI) programming environment. The MicroLabBox is capable of up to 24 channels of analog input and 16 channels of output in 16-bit at 1 Msps. There are also 48 channels of bidirectional digital inputs and outputs available. The MicroLabBox was ran at 10khz for these experiments although it is capable of up to 2Ghz processing in real-time. Force measurements were performed with an ATI Nano25 force transducer with NET F/T box interfaced with the dSPACE equipment using the ethernet protocol blocks in Simulink. A wiring schematic of the feedback system is shown in Fig. 5c. The force transducer F/T outputs are taken into the MicroLabBox and used to calculate forces on the test article. The motor driver is connected to two analog outputs, one for enabling the motor and the other for commanding the current of the motor. Another connection between the motor driver and MicroLabBox is made via the digital I/O ports to receive information from the motor encoder.

The goal of this experimental setup is to create a system that can employ different aerodynamic controllers for closed-loop feedback in unsteady aerodynamic applications. Given this goal, the system design consists of two individual feedback controllers. The first is a state feedback controller for the motor position itself. The aerodynamic control loop is then wrapped around the motor controller. This is shown in the block diagram in Fig. 6. The separation of these two controllers allows for different aerodynamic controllers to be designed that output the angle of attack based on the sensor feedback.

a)



b)



c)

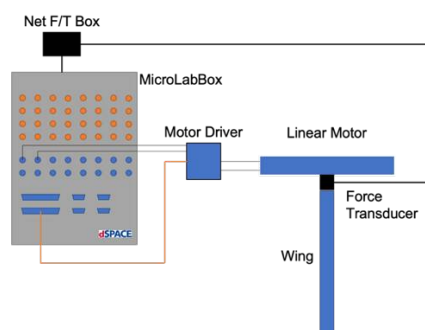


Fig. 5 a) University of Dayton water tunnel facility; b) CAD schematic of gust generator and test article; c) wiring schematic of MicroLabBox and input/output devices.

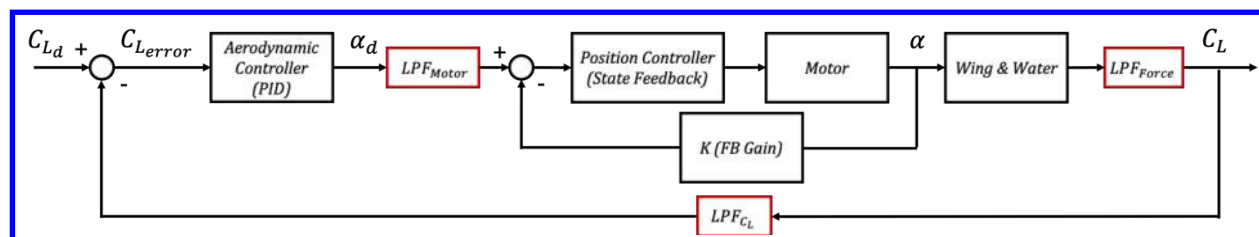


Fig. 6 Block diagram of system containing motor controller and lift controller.

III. Results

To understand the static performance of the flat plate, a static coefficient of lift curve was measured throughout the range of angles of attack for the attached and separated regimes. The plot of these is shown in Fig. 7. The results show that in the range of -30 to 30 degrees angle of attack the static flat plate achieves a maximum coefficient of lift of roughly 0.8 and a linear lift curve from -11 to 11 degrees beyond which the lift plateaus and the standard deviation increases.

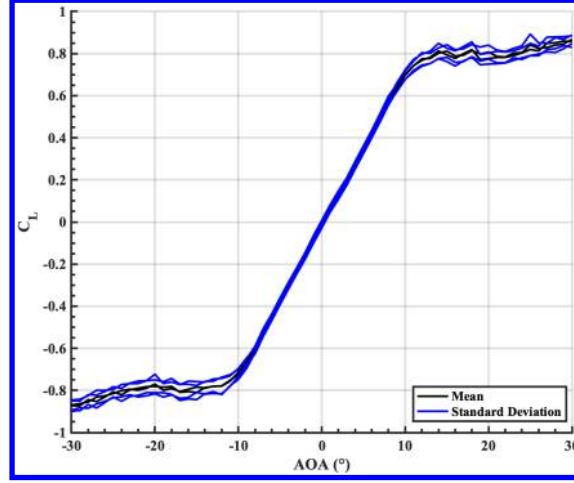


Fig. 7 Static coefficient of lift to angle of attack results.

In order to study a range of motions that encompass each range of reduced frequency from quasi steady to highly unsteady, the flat plate was pitched sinusoidally through the same range of angles of attack at 0.05, 0.02, 0.1, 0.2, and 0.3 reduced frequencies with the quasi-steady regime being $0 \leq k \leq 0.05$, the unsteady regime being $0.05 < k < 0.2$, and the highly unsteady regime being $0.2 \leq k$ [8]. Two representative cases of $k = 0.02$ and $k = 0.3$ are shown in Fig. 8 where the angle of attack that was commanded is plotted along with the coefficient of lift. It is clear from this data that the coefficient of lift response resembles the sinusoidal input better at the higher reduced frequency for the range of angles of attack of -30 to 30 that were tested. In the $k = 0.02$ case it is clear that the flow separation and stall effects of the plate dominate the response and it deviates from the sinusoidal input. In the $k = 0.3$ case these effects are not as dominant and the coefficient of lift response is much more sinusoidal.

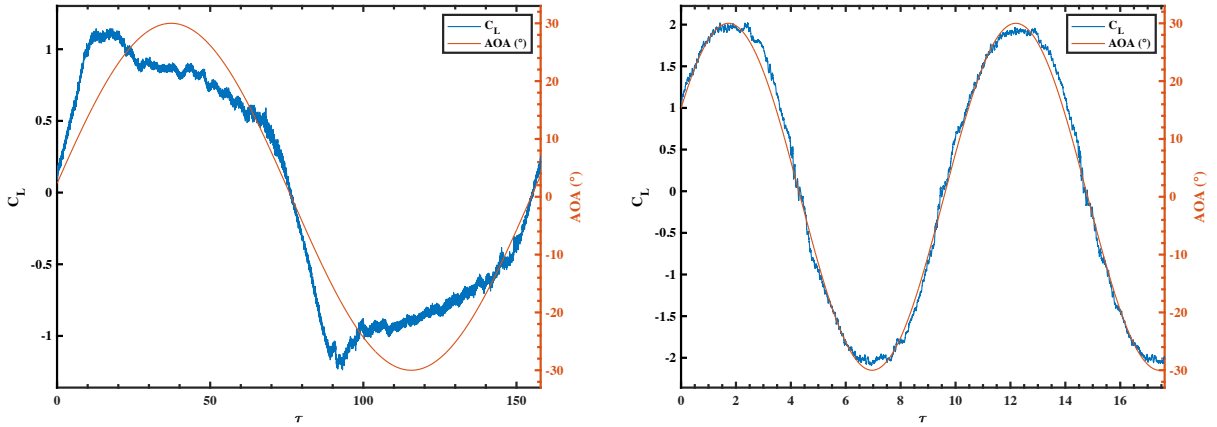


Fig. 8 Open loop alpha command angle of attack and coefficient of lift response for $k = 0.02$ (left) and $k = 0.3$ (right).

The plot of the coefficient of lift response of the flat plate through this open loop alpha control motion along with the static response is shown in Fig. 9 as well as the Theodorsen model prediction of the coefficient of lift. While these ranges of angles of attack and reduced frequencies are beyond the intended use of Theodorsen's model, it is

interesting to see similar trends when compared to the open loop response. The model underpredicted the overall lift coefficient achieved as expected but the general trends of changes in lift coefficient with reduced frequency matches well with the experiment. The coefficient of lift predictions as a function of angle of attack at different frequencies from Theodorsen's model seem to align well with the static lift-curve slope of the plate. The main changes are the offset in coefficient of lift magnitude at each angle of attack with increase in reduced frequency. The open loop experiments show the exact same behavior where the generic slope of all reduced frequency cases seem to align well with the static lift-curve slope of the plate. The alignment with the lift-curve slope seem to increase with increase in reduced frequency. One of the major differences between the experiment and the Theodorsen model arises in the post-stall region where massive flow separation takes place. As the plate moves through the highly separated region at a reduced frequency of 0.02 and 0.05, it maintains a similar coefficient of lift magnitude over a wide range of angle of attack (15° to 25°) which results in a slightly skewed and asymmetric hysteresis loop. This asymmetry in coefficient of lift magnitude in pitch up and pitch down maneuver seems to diminish at higher reduced frequencies making it more symmetric and aligned with the static-lift curve slope of the plate.

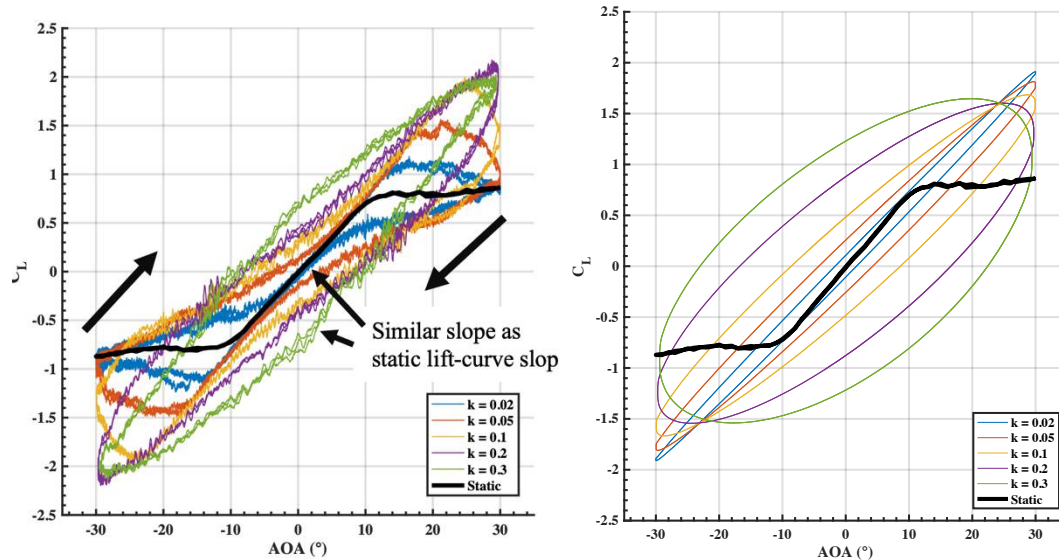


Fig. 9 Lift response from sinusoidal angle of attack sweep at different reduced frequencies (left) and Theodorsen's prediction (right).

The test matrix for the closed loop experiments that encompassed each of the reduced frequencies from the open loop tests is shown in Table 1 along with the desired coefficient of lift amplitudes that were feasible for each reduced frequency. Given that the onset of flow separation and stall is expected to limit the lower reduced frequency cases, their amplitudes were limited from the entire set. Each of these test points were run for 2 complete cycles of motion at their given reduced frequency and coefficient of lift amplitude.

Table 1. Closed loop test matrix.

Reduced Frequency (k)	Flow Type	2 Cycle Time	Feasible C_L Amplitude
0.01 [0.01 Hz]	Quasi-Steady	213 s	$\pm 0.2, 0.6, 0.8, 1.0$
0.05 [0.05 Hz]	Quasi-Steady	43 s	$\pm 0.2, 0.6, 0.8, 1.0, 1.2$
0.1 [0.1 Hz]	Unsteady	22 s	$\pm 0.2, 0.6, 0.8, 1.0, 1.2, 1.5, 1.8$
0.15 [0.14 Hz]	Unsteady	15 s	$\pm 0.2, 0.6, 0.8, 1.0, 1.2, 1.5, 1.8, 2.0, 2.2, 2.5$
0.2 [0.19 Hz]	Unsteady	11 s	
0.3 [0.29 Hz]	Highly Unsteady	8 s	$\pm 0.2, 0.6, 0.8, 1.0, 1.2, 1.5, 1.8, 2.0, 2.2, 2.5, 2.8$

A. PID Controller:

The PID controller used in this study was similar to the controller from Mongin and Gunasekaran [7] but with improved tuning done using step inputs and obtaining more appropriate gains. The gains that were used for the system are listed in Table 2.

Table 2. PID gain settings.

Controller Term	Gain
Proportional	1
Integral	5.5
Derivative	0.01

A test for how well the system tracked to the commanded value was performed by subtracting the root mean squared (RMS) error of the coefficient of lift response of the plate from the RMS of the desired coefficient of lift profile of the plate, the results of this are shown in Table 3 for each case. It is apparent that as the reduced frequency and amplitude are increased, higher error is found in the tracking of the controller. Shown in Fig. 10 is the closed loop tracking of the coefficient of lift displayed along with the resulting angle of attack of the system for tracking a coefficient of lift of 0.8 at both 0.01 and 0.3 reduced frequencies. The C_L Target signal is the desired output of the system and the C_L Filtered is the measured output of the system. The Motor Control Input signal is the output of the aerodynamic controller which is effectively the input to the motor controller and the AOA Sensed signal is the measured position of the flat plate in degrees or angle of attack. These two cases show the comparison of the same coefficient of lift amplitude but the higher reduced frequency of 0.3 there is a phase lag and a slight undershoot of the target coefficient of lift profile explaining the RMS error increase from -0.0003 to 0.02228.

Table 3. PID Root Mean Squared Error.

		CL Amplitude										
		0.2	0.6	0.8	1	1.2	1.5	1.8	2	2.2	2.5	2.8
RF	0.01	-0.0001	0.0000	-0.0003	0.0035							
	0.05	-0.0006	-0.0001	0.0011	0.0071	0.0238						
	0.1	-0.0010	0.0036	0.0086	0.0066	0.0214	0.0481	0.0787				
	0.15	-0.0004	0.0029	0.0001	0.0061	0.0235	0.0438	0.0696	0.0784	0.0742	0.1639	
	0.2	0.0047	0.0082	0.0035	0.0149	0.0213	0.0217	0.0530	0.0479	0.0830	0.1585	
	0.3	0.0082	0.0155	0.0228	0.0131	0.0195	0.0212	0.0578	0.0491	0.0864	0.1177	0.2049

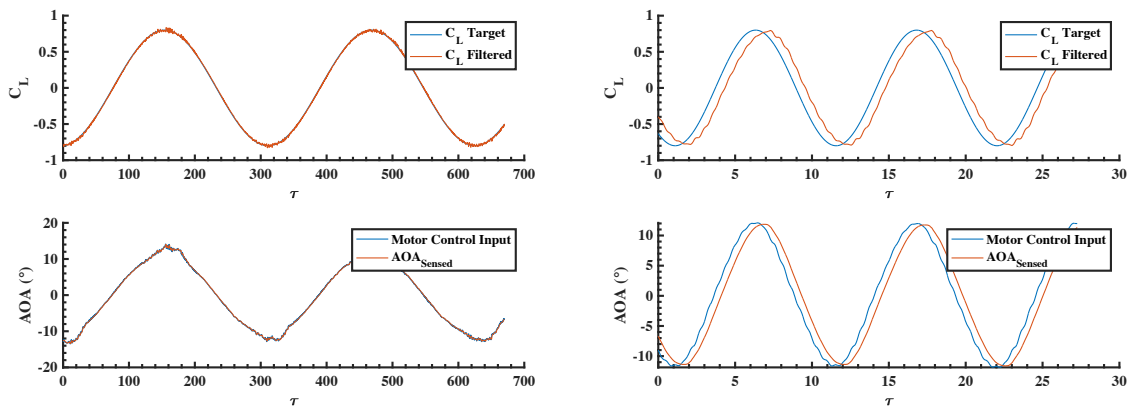


Fig. 10 Closed Loop C_L and AOA amplitude tracking for $C_L=0.8$ at $k=0.01$ (left) and $k=0.3$ (right).

The response of the PID based closed loop control system was studied to better understand the system's bandwidth and characteristics at different operating points. This was done by comparing the phase shift and magnitude change between input and output signal for each reduced frequency and amplitude cases. The Bode plot of these results is shown in Fig. 11 for each of the coefficient of lift amplitude cases. The system shows a good magnitude response up to the reduced frequency of 0.3 where all of the cases show a significant decrease in magnitude from 0.2. The system also appears to have significant phase lag after reduced frequencies of 0.15 are reached. This is again in alignment with the RMS error analysis from earlier.

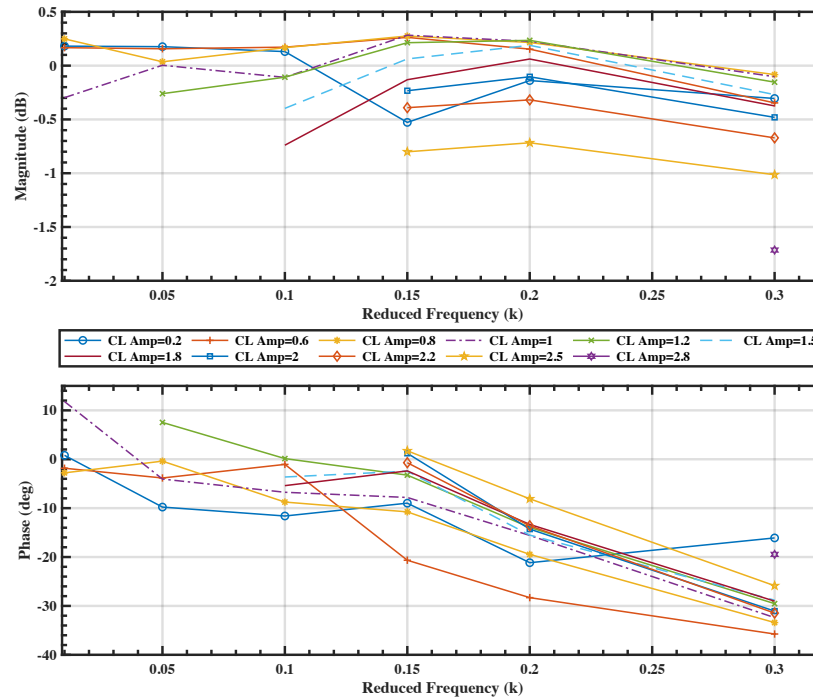


Fig. 11 Closed Loop PID Controller Bode Plot.

The results from the PID based closed loop control are shown in Fig. 12 in the same style as the open loop alpha sweep experiments in Fig. 9 to compare it with the static lift-curve slope. There are clear trends that arise in these plots as reduced frequency is increased. At the reduced frequency of 0.01, the closed loop control response to sinusoidal coefficient of lift input seems to track extremely well with the static coefficient of lift curve slope. The hysteresis are only observed at higher angles of attack where the flow is fully separated. Increasing the reduced frequency to 0.05, slight differences can be seen in the coefficient of lift variation with α where the pitch up and pitch down motion of the lift curve have a similar slope as the static lift-curve slope. At the transition region between pitch up and pitch down, the plate experiences a highly separated flow where similar coefficient of lift magnitude is maintained over an extended angle of attack range. This trend becomes a lot more explicit at the reduced frequency of 0.1. It should be noted that all coefficient of lift inputs at different reduced frequencies are sinusoidal. Therefore, any change in the trends or the hysteresis loops comes from the changes in the angle of attack that the controller feeds in to successfully track a given coefficient of lift profile. At k of 0.1, the plate was able to track at a maximum coefficient of lift amplitude of 1.8, which is a lot higher than the maximum lift achieved in the static case. The coefficient of lift amplitude of 1.8 is achieved at an angle of attack of 20° , however the plate “rests” at this coefficient of lift magnitude while the angle of attack increases to 35° and then starts to decrease in coefficient of lift. Several undulations are seen in the pitch down and pitch up route the plate takes to track coefficient of lift. This could be due to the controller actively trying to match the coefficient of lift at every time step. Even with the undulations, the general slope of the lift curve loop remains like that of the static-lift curve slope. At k of 0.15, the same trends can be seen for all coefficient of lift amplitude cases. However, the “resting” angles of attack where the plate maintains the maximum coefficient of lift amplitude are significantly decreased. A sharp drop off in coefficient of lift magnitude can be seen at the same angle of attack in the transition region between pitch up and pitch down. The undulations that are seen in

the pitch up and pitch down route at k of 0.1 occur with less frequency at k of 0.15 and the generic slope of the hysteresis loop still matches to that of the static lift-curve slope. At k of 0.2, the sharp drop off in coefficient of lift magnitude at higher angles of attack is reduced to a moderate range of coefficient of lift. The pitch up and pitch down route taken by all the coefficient of lift amplitudes overlap on top of each other and the curves start to deviate from each other only at the transition regions between pitch up and down maneuvers and vice-versa. This overlap of the coefficient of lift hysteresis loops at the pitch up and pitch down phase of the motion breaks down at the k of 0.3 where each coefficient of lift amplitude is different and occurs at a slightly elevated coefficient of lift magnitudes with increase in the input coefficient of lift amplitude. One common theme in all the reduced frequency cases is that the generic slope of the hysteresis loops follow a similar slope to that of the linear portion of the static lift-curve.

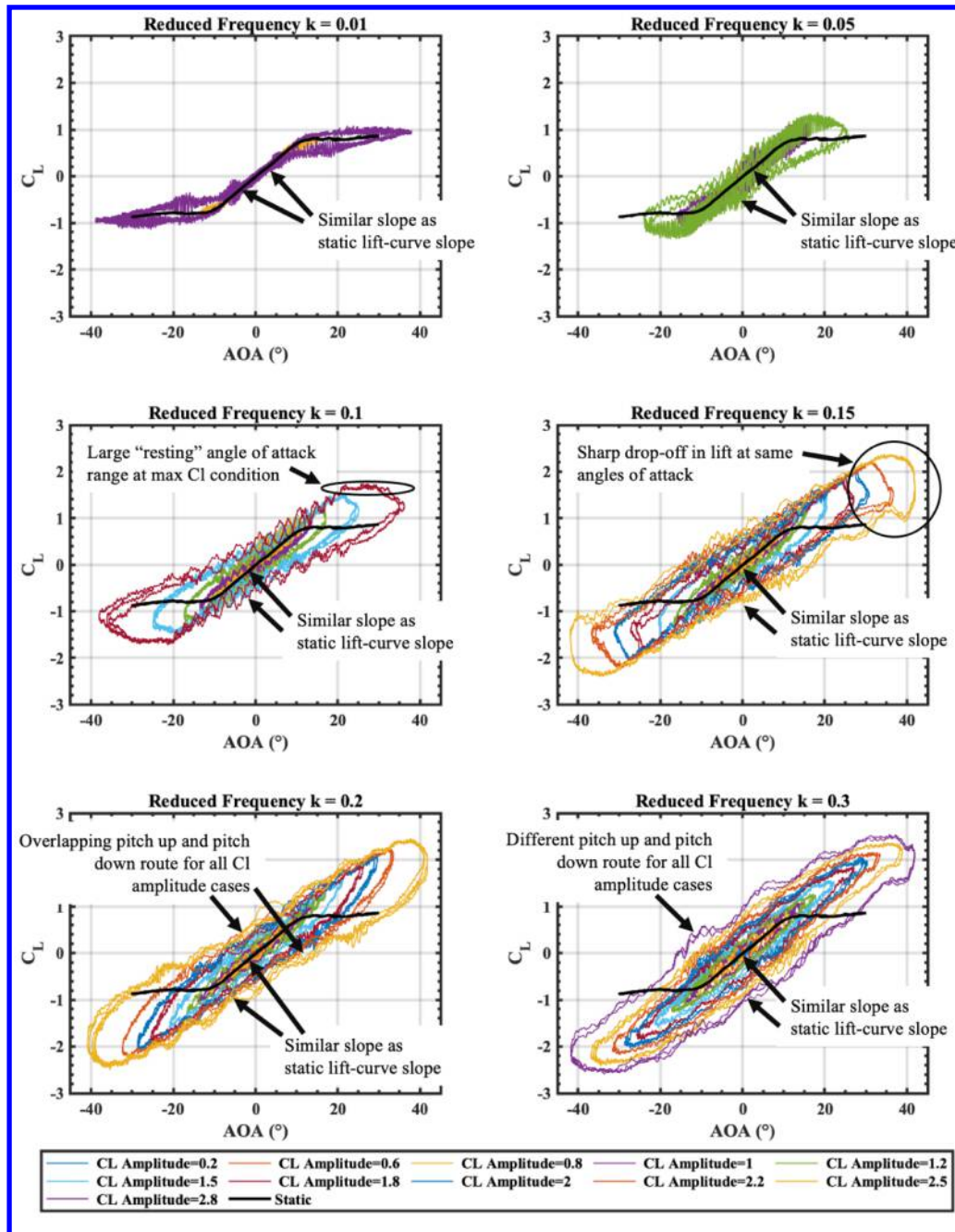


Fig. 12 Closed loop angle of attack response of PID controller for each reduced frequency.

B. Full State Feedback Controller:

From the system identification process described in the introduction section, the model in Equation 9 of the open loop system was obtained from the step response shown in Fig. 13.

$$\begin{aligned}\dot{x} &= Ax + Bu = \begin{bmatrix} -5.6 & -33.8 \\ 32 & 0 \end{bmatrix} x + \begin{bmatrix} 4 \\ 0 \end{bmatrix} u \\ y &= Cx + Du = \begin{bmatrix} 0 & 6.6 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \end{bmatrix} u\end{aligned}\quad (9)$$

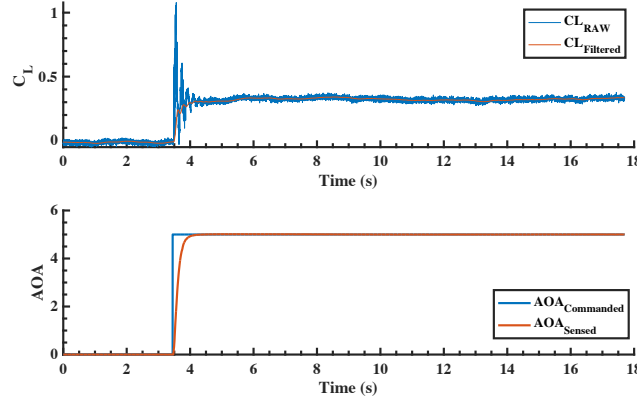


Fig. 13 Open loop step input and response plot.

The transfer function of the desired system response was tuned to have $\zeta = 0.99$ and $\omega_0 = 31.5$ resulting in the transfer function shown in Equation 10. The resulting desired poles are shown in Equation 11.

$$TF_{Desried} = \frac{992.25}{s^2 + 62.37s + 992.25} \quad (10)$$

$$P_{Desried} = [-31.1 + 4.4i; -31.1 - 4.4i] \quad (11)$$

From this, the resulting closed loop gains were found as shown in Equation 12. A DC gain of 0.766 was used to achieve the desired magnitude response and was used to calculate the reference gain K_R needed to properly scale the system, as shown in Equation 13.

$$K = [14.14, -0.75] \quad (12)$$

$$K_R = \frac{1}{K_{DC}} = 1.18. \quad (13)$$

Once the system was tuned the same set of tests were attempted on the flat plate with the full state feedback controller as were with the PID controller. The difference in RMS error of the system is shown in Table 4 for each of the cases run where a similar trend to the PID controller is shown where with higher reduced frequency there is higher error and with higher coefficient of lift amplitudes error increases as well. This again leads to the need for a closer analysis of the phase and magnitude response of the system. Looking at the cases of coefficient of lift magnitude 0.8 for reduced frequencies of 0.01 and 0.3 (same cases as Fig. 10 for PID) in Fig. 14 for the state feedback controller a similar phase lag is present at the higher reduced frequency but the magnitude is no longer smaller, rather the lower reduced frequency case had more error in this test than the higher reduced frequency with 0.006 and 0.005 respectively. This is a small error but overall this shows that the two controllers are operating differently.

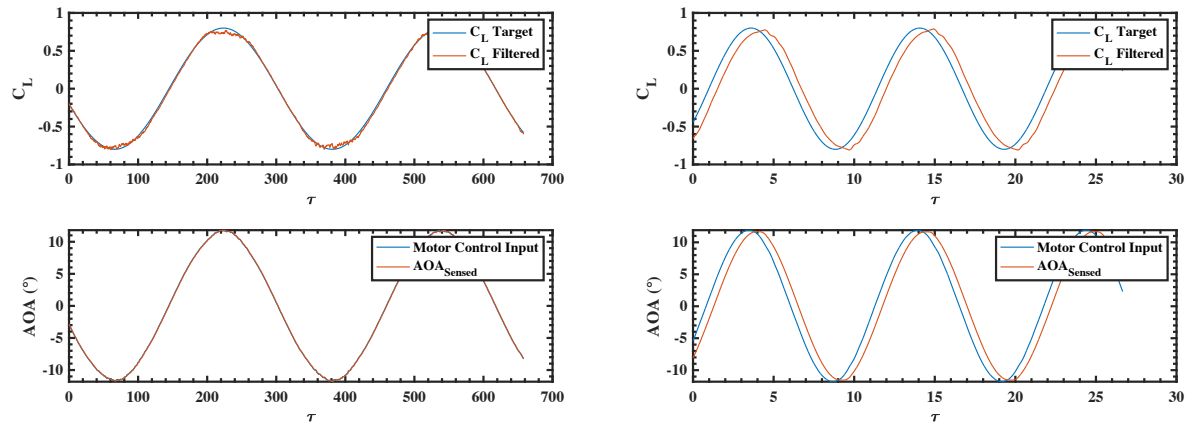


Fig. 14 Closed Loop C_L and AOA amplitude tracking for $C_L=0.8$ at $k=0.01$ (left) and $k=0.3$ (right).

Table 4. State Feedback Controller Root Mean Squared Error.

		CL Amplitude										
		0.2	0.6	0.8	1	1.2	1.5	1.8	2	2.2	2.5	2.8
RF	0.01	0.001	-0.010	0.006	0.083							
	0.05	0.002	-0.013	-0.004	0.051	0.122						
	0.1	0.002	-0.013	-0.021	0.002	0.047	0.109	0.197				
	0.15	0.004	-0.006	-0.018	-0.029	-0.004	0.050	0.103	0.136	0.196	0.291	
	0.2	0.004	0.002	-0.027	-0.021	-0.014	0.004	0.058	0.079	0.126	0.243	0.377
	0.3	0.008	0.018	0.005	0.009	0.009	0.011	0.022	0.037	0.067	0.116	0.245

Fig. 15 shows the Bode plot of the closed loop state feedback controller where the magnitude is shown to increase up to a reduced frequency of 0.15 and decrease rapidly after a reduced frequency of 0.2. This shows the controllers inability to track to the full magnitude of the profile such as in the cases of amplitudes of 2.5 and 2.8 where the magnitudes are consistently -1 and -1.5 decibels respectively. These same cases have some of the best phase alignment, this was the same case as the PID controller. This indicates that while the controller is unable to keep up with the magnitude being requested at these highest cases, it is staying close to in phase with the commanded signal. The phase lag of the system with the state feedback controller gradually decreases with increase in frequency when compared to the abrupt decrease in the case of the PID controller.

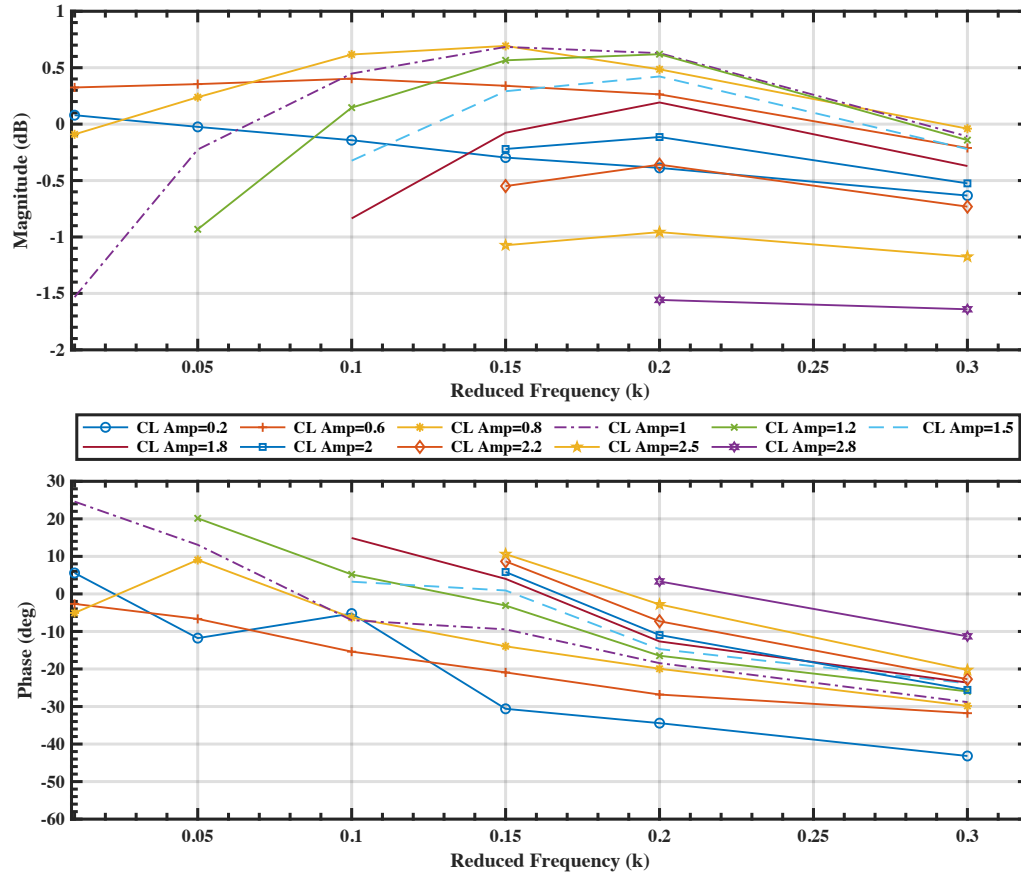


Fig. 15 Closed Loop State Feedback Controller Bode Plot.

The plots of each reduced frequency case with each coefficient of lift amplitude are displayed as coefficient of lift versus angle of attack in Fig. 16. The behavior of the coefficient of lift response in these cases is extremely similar to that of the PID controller from Fig. 12. In each of the reduced frequency cases we see that the trend in the profile that the coefficient of lift takes on the pitch up route of movement matches the slope of the static lift curve. The lower reduced frequency cases of 0.01 and 0.05 have less deviations in coefficient of lift near the transition to separation than the PID controller had. This is due to the state feedback controller not responding to the tracking error as quickly as the PID. The tracking of coefficient of lift in each of these cases is more compact to the static lift curve slope than the PID controller results as well. The PID controller moved to higher angles of attack ($\pm 30^\circ$) to achieve the lift tracking desired than the state feedback controller ($\pm 15^\circ$) in the 0.01 reduced frequency case. The “resting area” where coefficient of lift stays constant while angle of attack increases that was previously seen in the reduced frequency 0.1 case in the PID controller is shown to appear in the state feedback controller as well but with less range of angle of attack (20° to 25°). This makes the transition from pitch up to pitch down smoother as well. The reduced frequency 0.15 case exhibited a drop off in lift after 35° and no “resting area”. In both the 0.1 and 0.15 reduced frequency cases the state feedback controller exhibited less undulations in coefficient of lift throughout the pitch down movement than the PID controller. At a reduced frequency of 0.2 the pitch up routes overlap for each coefficient of lift amplitude and after the transition to pitch down deviate in response until the transition back to pitch up in the negative direction. At a reduced frequency of 0.3 the pitch up response of the coefficient of lift for each amplitude no longer overlap and they each take their own route around the hysteresis loop. It is interesting to note that at this highest reduced frequency the general shape of each path is extremely similar and seems as if it were scalable by coefficient of lift amplitude directly.

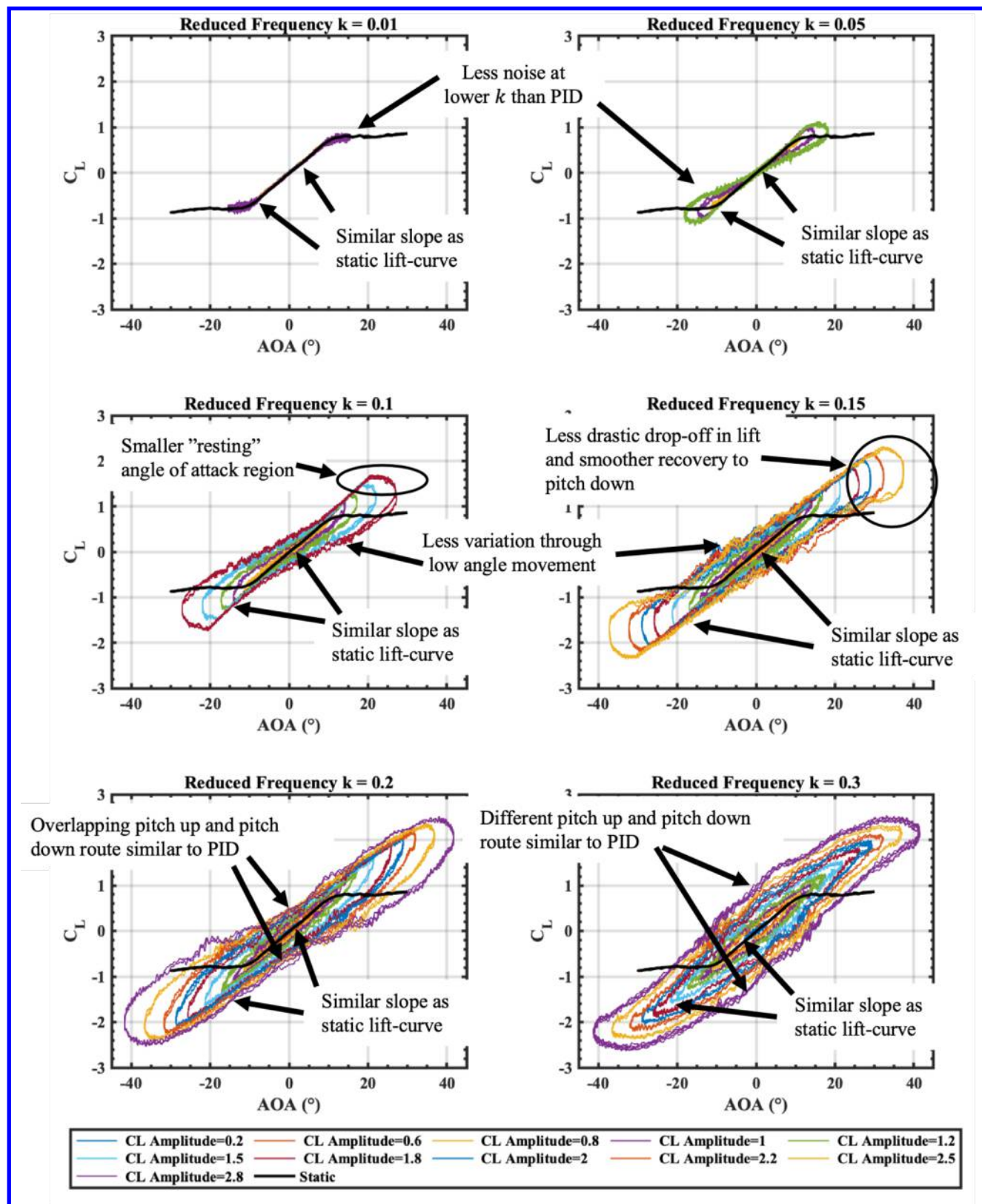


Fig. 16 Closed loop angle of attack response of the full state feedback controller for each reduced frequency.

IV. Analysis

A. Comparison between PID and SF Controllers

The two controllers achieved coefficient of lift tracking of the same desired coefficient of lift signal with slightly different angle of attack histories. Shown in Fig. 17 is the reduced frequency of 0.3 and coefficient of lift amplitude of 2.8 case which is the highest coefficient of lift amplitude and reduced frequency case tested. It can be clearly seen that both the controllers cannot achieve the desired coefficient of lift of 2.8 with the pitch rate. However, the angle of attack histories between the PID and the SF are a lot different except for the angle of attack magnitude. Both systems achieve a similar maximum angle of attack during these tests despite the system being free to move from -60 to 60 degrees. At the peak lift, the PID controller holds the same angle of attack for a longer period than the SF controller.

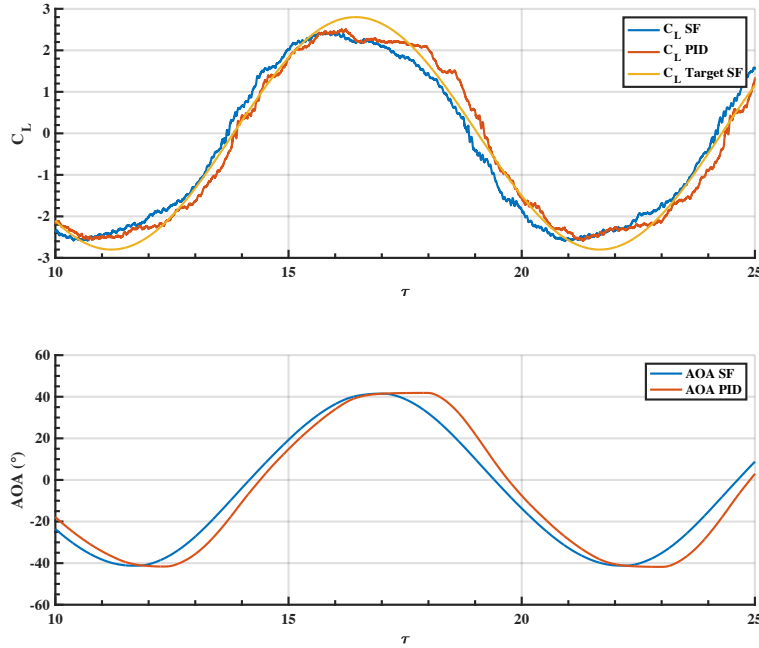


Fig. 17 Coefficient of lift and angle of attack history for the reduced frequency $k = 0.3$ with a C_L 2.8.

Similar behavior is seen for other reduced frequency cases as well. Fig. 18 shows the angle of attack history for two cases at a reduced frequency of $k = 0.2$ with coefficient of lift amplitudes of 1.8 and 2.0. The coefficient of lift tracking for each case compares well to the target however the angle of attack histories are slightly different from one another especially on the pitch down motion where the plate is going from higher angles of attack to lower angles of attack. This could be due to many different reasons including controller tuning for each case. Although the angle of attack histories took slightly different trajectories from each controller, the overall coefficient of lift profile for each case aligned similarly to the target coefficient of lift. However, it should be noted again that the PID controller offers better coefficient of lift amplitude tracking than the state feedback controller. Therefore, the angle of attack histories from the PID controller is more indicative of the angle of attack requirement to maintain coefficient of lift.

Another set of cases shown in Fig. 19 display the difference in the two controllers' angle of attack profiles in quasi-steady and highly unsteady cases. In k of 0.01 case where unsteady effects are minimal to none, the PID controller deviates greatly from the state feedback controller and attempts to achieve the desired coefficient of lift of 1 by continually increasing its angle of attack. This effect doesn't happen at k of 0.2. The difference between the response of the two controllers is likely due to the tuning parameters used for each of them. The PID controllers integral gain is quite high, giving the system much more bandwidth to react to an increase in error over time. The state feedback controller poles were not set to extremely rigid responses due to the pole placement technique used. This results in a PID controller that is very quick to respond to error but might also oscillate around a value due to overshoot

and a state feedback controller that is less likely to overshoot but might not achieve the full coefficient of lift magnitude desired. Each of these controllers responds and achieves the desired settling time of a step response however a higher order controller such as an adaptive controller could learn and adapt to the error of the system through time and arrive at a much more robust controller for all cases.

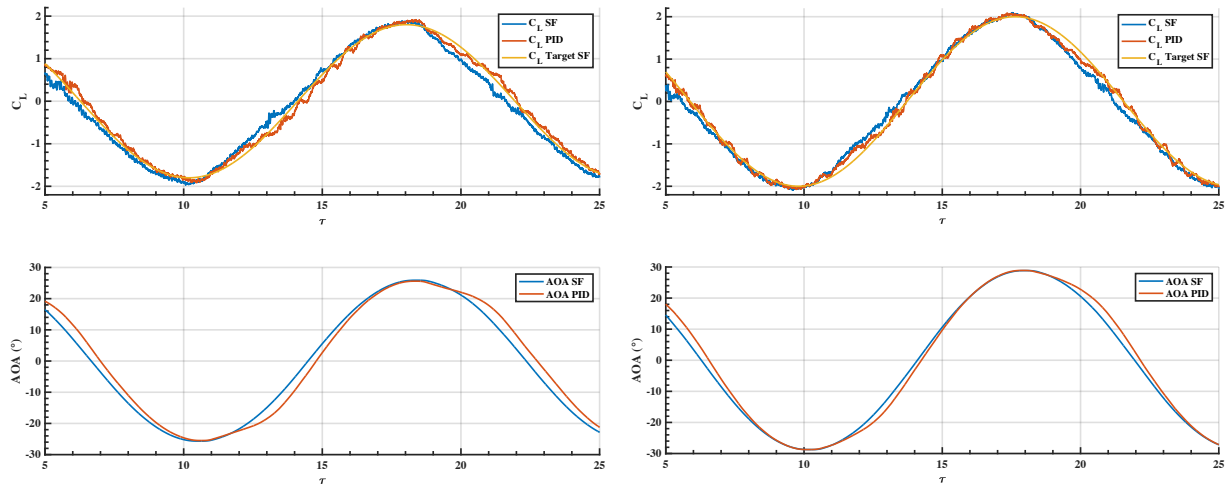


Fig. 18 Coefficient of lift and angle of attack history for the reduced frequency $k = 0.2$ with a C_L 1.8 (left) and C_L 2.0 (right) profile.

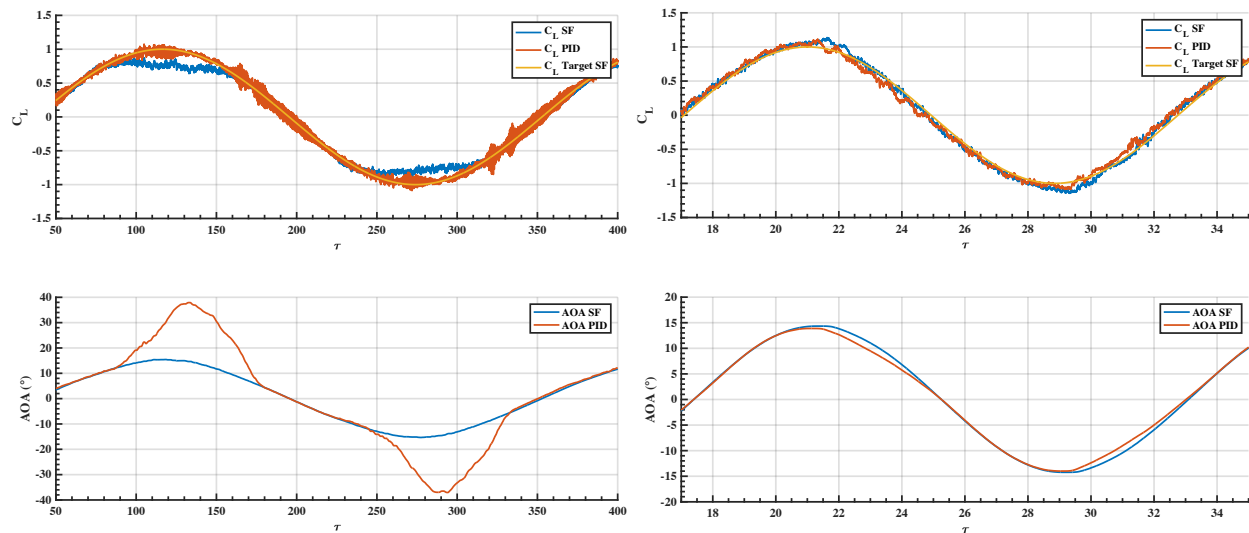


Fig. 19 Coefficient of lift and angle of attack history for cases of C_L 1 at reduced frequencies of $k = 0.01$ (left) and $k = 0.2$ (right).

B. Theodorsen's coefficient of lift predictions using angle of attack histories from SF controller

It is known that the Theodorsen's model is not quite applicable to high amplitude and high frequency cases where massive flow separation occurs. However, strong comparisons in the trends between Theodorsen's model and the closed loop control coefficient of lift vs alpha history can be observed. Using the angle of attack history of the full state feedback cases, Theodorsen's model was used to predict the coefficient of lift histories for each case and the results are shown in Fig. 20. Even though the angle of attack histories from the PID controller is a lot more representative of the angle of attack needed to track coefficient of lift, the SF controller provides a robust and a less noisy signal when compared to PID. Fig. 20 shows the limitations of the Theodorsen's model clearly. While the model makes a reasonable prediction at lower angles of attack, the flow dynamics in the regions where massive flow

separation occurs could not be captured by the model. The generic slope of the hysteresis loop while being similar to the static lift-curve slope as in the case of closed loop experiments, the results in Fig. 20 shows that at higher reduced frequencies, the general slope of the coefficient of lift orbit is slightly higher than that of the static lift curve. However, given the global resemblance of the lift and angle of attack histories between the Theodorsen and closed loop control experiments, the model can be adopted in the future for better predictions.

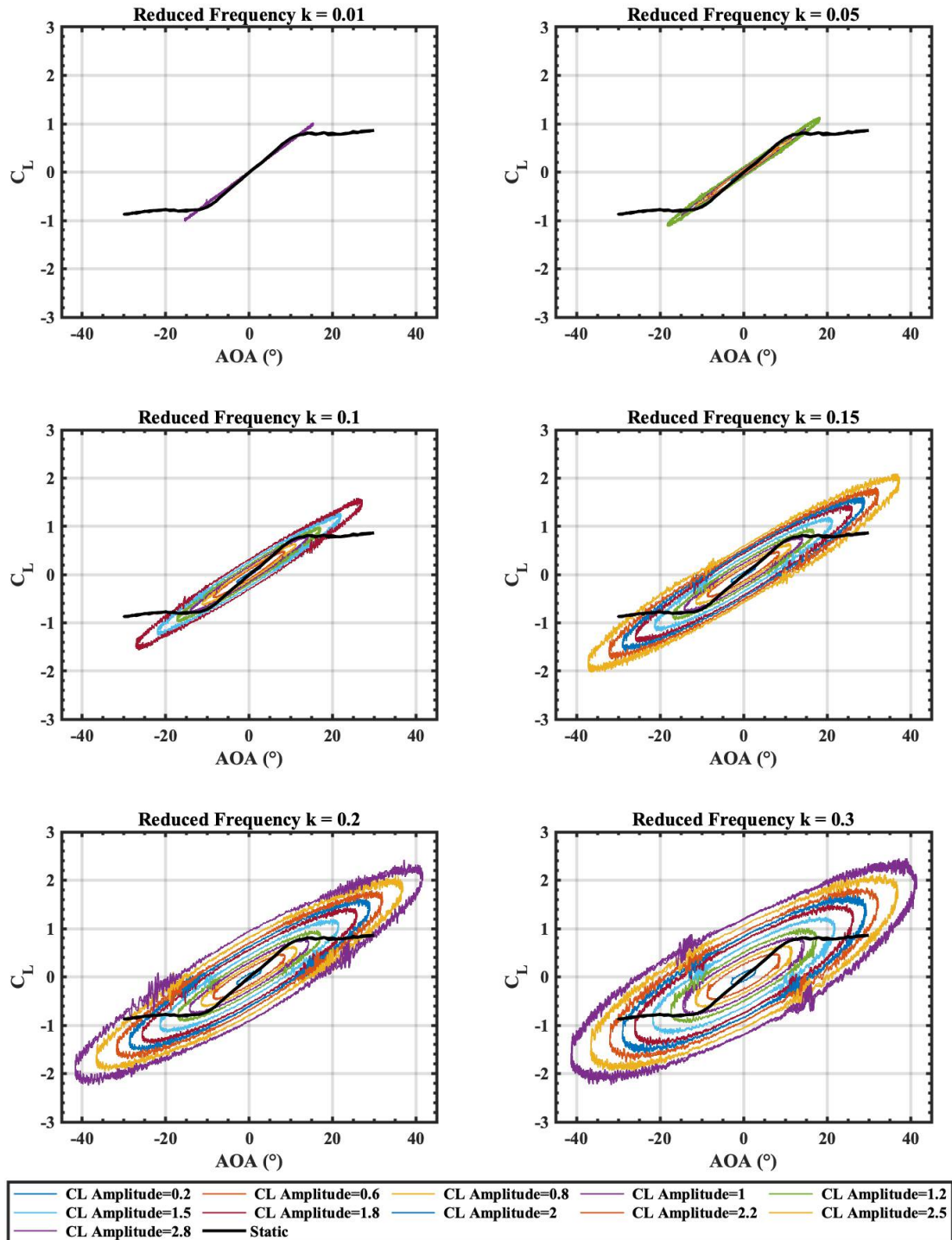


Fig. 20 Theodorsen predicted coefficient of lift from closed loop state feedback angle of attack history.

A direct comparison is made between the Theodorsen's CI prediction based on angle of attack history from the SF controller and the experimental data for k of 0.2 case in Fig. 21. The magnitude and the trends predicted by Theodorsen's model align well with the experimental data and is common across all the reduced frequency cases, except for the highly non-linear behaviors. With an adaptive controller architecture, the Theodorsen model could be used to inform the controller of the general dynamics of the system while the adaptations laws could then be implemented to correct for the error. This would provide a system that could adapt and learn a model that would work for these types of experiments and likely beyond the range of reduced frequencies and coefficient of lift amplitudes tested here. In addition to the Theodorsen model, other higher order aerodynamic models such as the G-K model could be used for this purpose and compared.

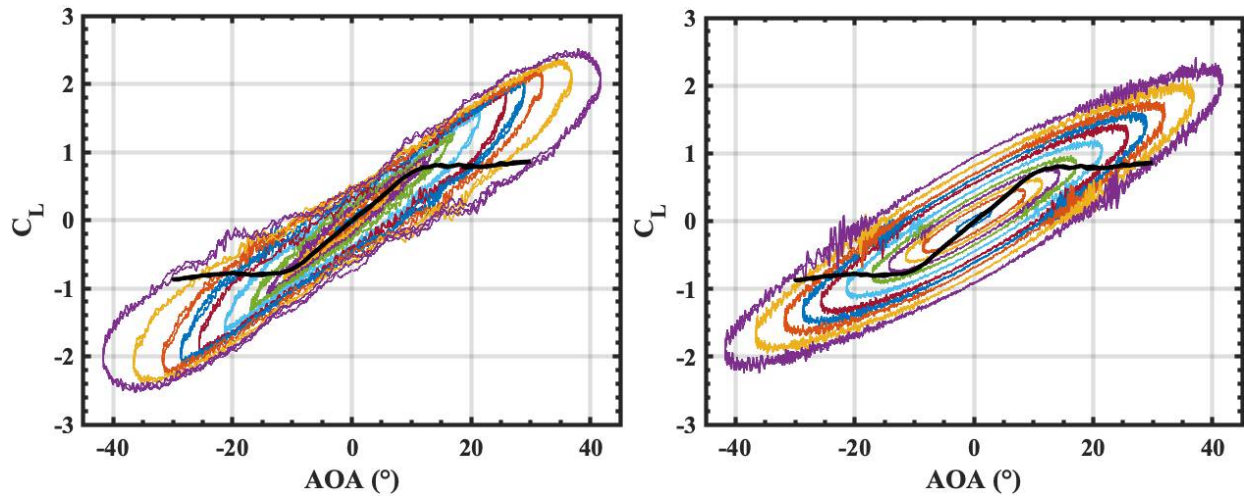


Fig. 21 Experimental coefficient of lift and Theodorsen predicted coefficient of lift response for $k = 0.2$.

V. Conclusions

The open loop coefficient of lift response of a pitching flat plate at high reduced frequencies and high angles of attack has been well understood for many years. This study allowed a view into the angle of attack history of a closed loop coefficient of lift tracked flat plate and the resulting trends of the system. In analyzing the angle of attack data of both a PID and state feedback controller, the underlying dynamics associated with high amplitude lift tracking have been seen. The controllers each displayed a linear trend to the lift curve slope when the plate was traveling in a pitch up direction. In the unsteady regime of reduced frequencies of 0.1 and 0.15 the controller formed “resting areas” where it continued to pitch the plate despite the coefficient of lift maintaining a constant value but resulting in accurate coefficient of lift tracking of the system. At a reduced frequency of 0.3 the coefficient of lift response of the system formed a general shape that maintained form and scaled based on coefficient of lift amplitude. These findings confirm that controllability of a flat plate in regimes beyond the point of flow separation is possible with closed-loop controllers of the PID and full state feedback types up to reduced frequencies of 0.3 and coefficient of lift amplitudes of 2.5. The PID and full state feedback controllers each performed slightly different angle of attack trajectories to achieve similar coefficient of lift profiles at lower reduced frequencies. However, at higher reduced frequencies the two angle of attack profiles from the controllers were similar. When each controller was given the maximum reduced frequency of 0.3 and coefficient of lift of 2.8 tracking signal, the controllers performed slightly different angle of attack profiles to attempt to achieve the desired coefficient of lift which was not met by either controller. The comparison of the experimental data from the state feedback controller to a Theodorsen model prediction using the angle of attack history of the closed loop state feedback controller yielded a prediction that resembles the experimental data close enough to merit further investigation of adaptive controller implementation.

VI. Future work

Using the Theodorsen model and the known state feedback controller of the system used in these experiments, an adaptive controller could be designed to adapt to the physical plant using the Theodorsen model as a prediction of the dynamics and an approximator to adjust the controller based on tracking error. This controller could potentially improve the phase and magnitude response of the system, yielding much better tracking to the target coefficient of lift and potentially expand the bandwidth of reduced frequencies that the system can handle. The approximated plant of the system that the adaptive controller produces could also be extremely useful in modeling and understanding the dynamics of higher reduced frequency movements at these higher coefficients of lift. Other models such as the G-K model could be used to inform adaptive controllers as well.

VII. References

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